

Daniel Afkhami

(732) 694-9968 | dan.afkhami@gmail.com | linkedin.com/in/danielafkhami | github.com/danielafkhami

EDUCATION

Rutgers University - Honors College	Piscataway, NJ
<i>Bachelor of Science in Computer Science and Mathematics GPA: 3.94/4.00 (Dean's List)</i>	<i>Expected May 2029</i>
• Coursework: Systems Programming, Data Structures, Computer Architecture, Mathematical Theory of Probability, Real Analysis, Elementary Differential Equations, Linear Algebra • Activities: Rutgers Symphonic Winds, ColorStack, Rutgers University Ski Team, Rutgers Cars	
Metuchen High School	Metuchen, NJ
<i>GPA: 4.51/4.00 (Summa Cum Laude) SAT: 1540 (800 Math, 740 Reading & Writing)</i>	<i>Graduated June 2025</i>

EXPERIENCE

Software Engineer Intern	May 2026 - Present
<i>Stone Temple Partners, LLC</i>	<i>New York, NY</i>
Full-stack engineer on a wealth-management platform for RIAs serving UHNW households.	
• Implemented CRM functionality in use by 8 advisors overseeing \$750M+ AUM, designing RESTful endpoints using FastAPI for a React/TypeScript frontend backed by a Supabase/PostgreSQL schema. • Shipped a portfolio analysis and tax-aware rebalancing engine that scores a prospective client's portfolio against a Monte Carlo-optimized target and generates executable transition plans with capital-gains netting, tax-loss harvesting, and wash-sale constraints. • Delivered a Windows desktop application (PySide6) that records advisor calls, transcribes them via Azure AI Speech, and files LLM-generated summaries to five CRM record types, cutting transcription costs ~95% by retiring a third-party service. • Built automated ETL pipelines (Supabase Edge Functions on pg_cron) that reconcile position-level data via the Addepar API, giving advisors direct access to household AUM, cash balances, and per-fund performance for pooled vehicles. • Replaced password authentication with Microsoft OAuth 2.0 SSO and built an in-app mail client via the Microsoft Graph API with automated email filing, eliminating stored credentials and routing every Microsoft integration through a single authentication flow.	

PROJECTS

D3PM Synthesizing 2D Ising System Configurations	<i>April 2026 - September 2026</i>
<i>Python, PyTorch, NumPy, SciPy, Numba, Matplotlib</i>	
• Built a Numba-compiled Metropolis Monte Carlo sampler to generate a dataset of 32×32 Ising model lattice states at the critical temperature. • Designed a 300-step PyTorch Discrete Denoising Diffusion Probabilistic Model (D3PM) using a U-Net architecture, exact binary reverse transitions, and EMA-stabilized refinement. • Evaluated 500 generated configurations against 500 held-out simulations by comparing energy and magnetization distributions with KS and Wasserstein statistics, demonstrating strong overall agreement with the reference data.	
PINN Modeling Field-Induced Motion of Ferrofluid	<i>August 2024 - September 2025</i>
<i>Python, TensorFlow, NumPy, SciPy, Matplotlib</i>	
• Derived the governing ODE balancing nonlinear magnetic force with Stokes drag to model the movement of a ferrofluid droplet through a viscous medium under a magnetic-field gradient. • Built a TensorFlow Physics-Informed Neural Network (PINN) trained on 4 noisy observations and 1,000 collocation points, combining data loss with an automatically differentiated physics residual. • Compared predicted traversal times with a SciPy quadrature reference over 0-20 mm, achieving accurate extrapolation beyond the 15-20 mm observation interval.	

SKILLS

Languages: Python, Java, C, C++, TypeScript, JavaScript, SQL, HTML/CSS, Bash
Frameworks & Libraries: React, FastAPI, Tailwind CSS, PyTorch, TensorFlow, NumPy, SciPy, PySide6
Data & Cloud: PostgreSQL, Supabase, Azure
Tools & Platforms: Git, Docker, Vite, Vercel, Railway, Addepar